Evaluation impact of increased TCP on name servers

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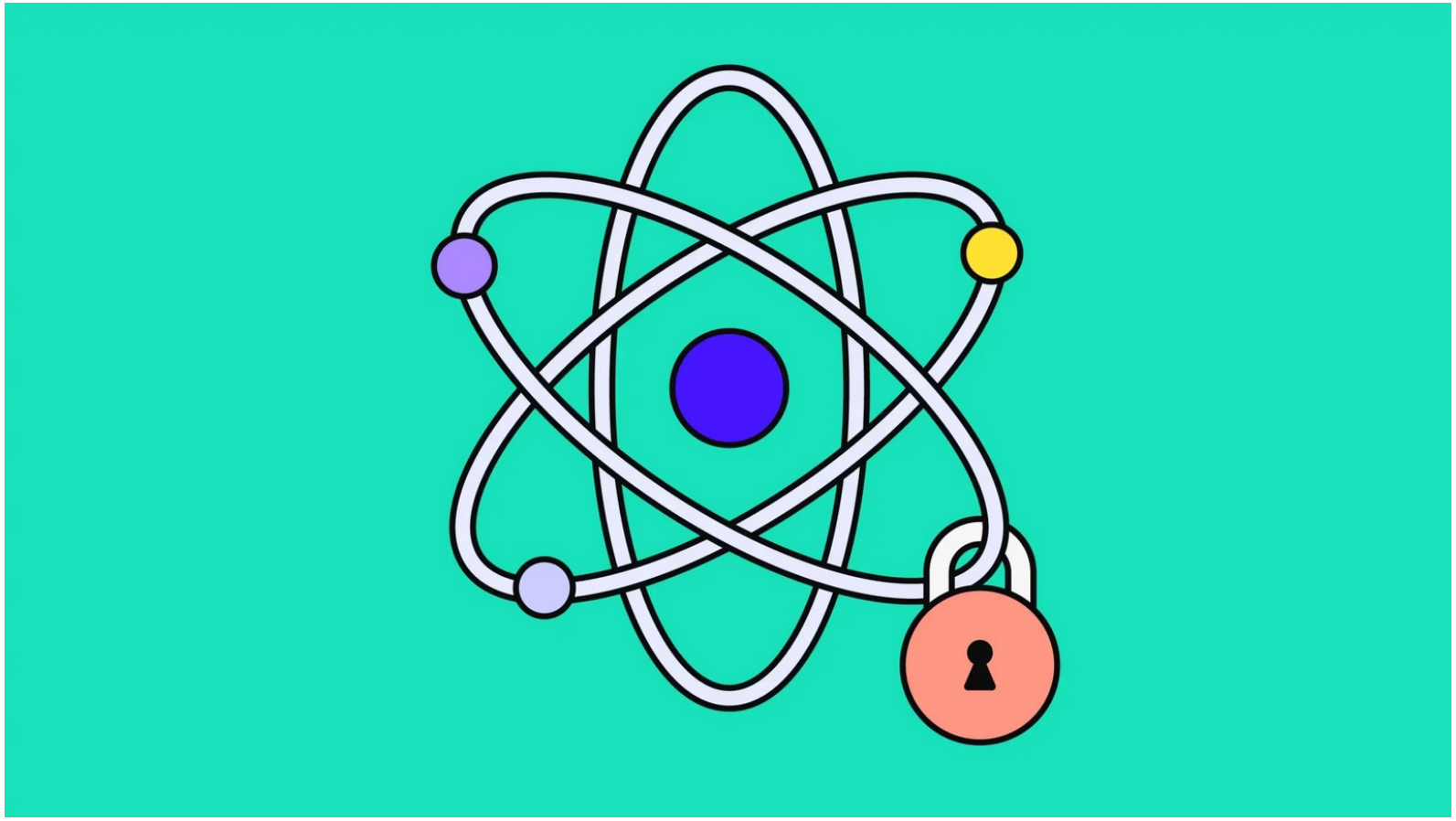
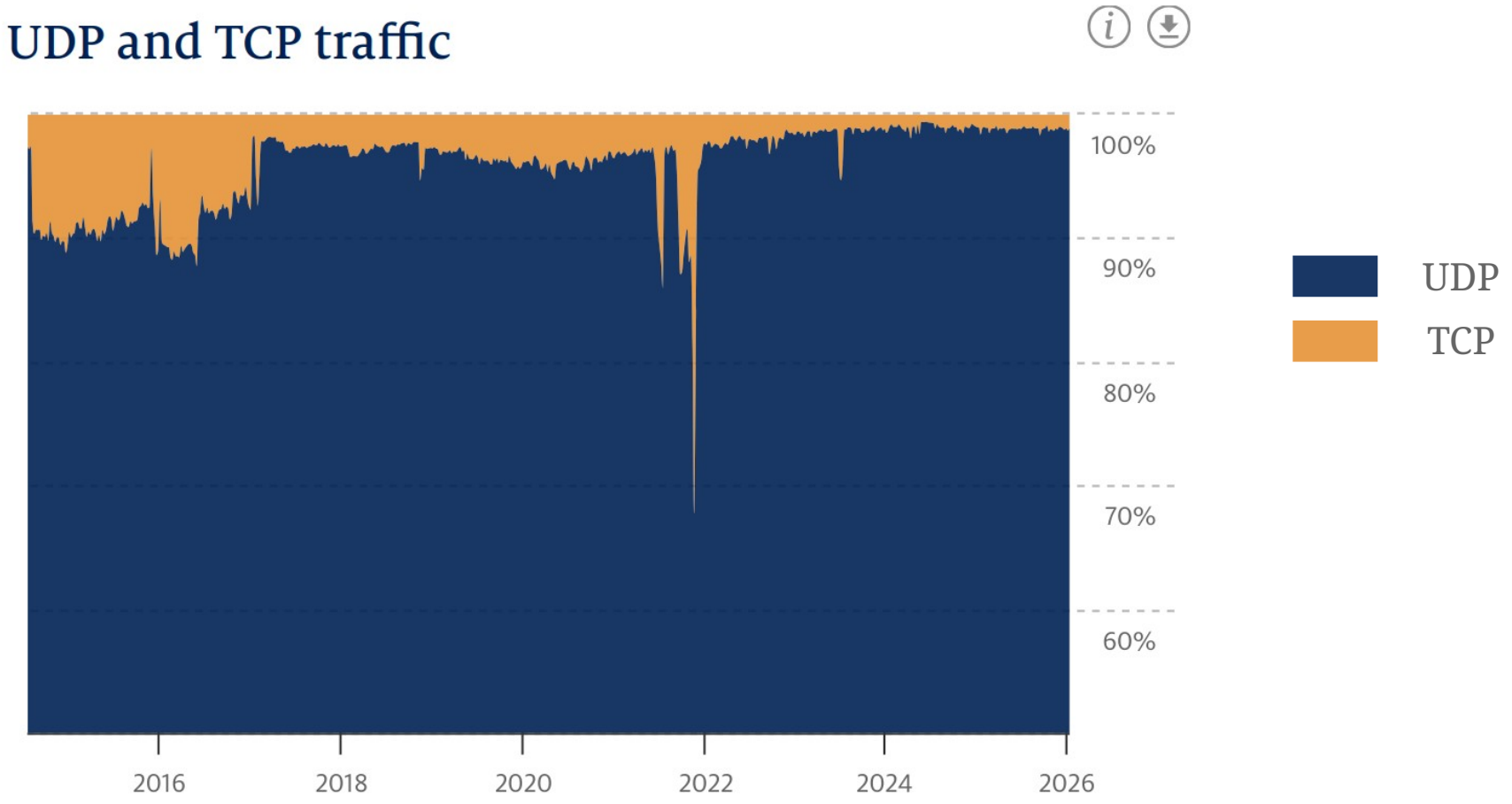
DNSSEC and PQC

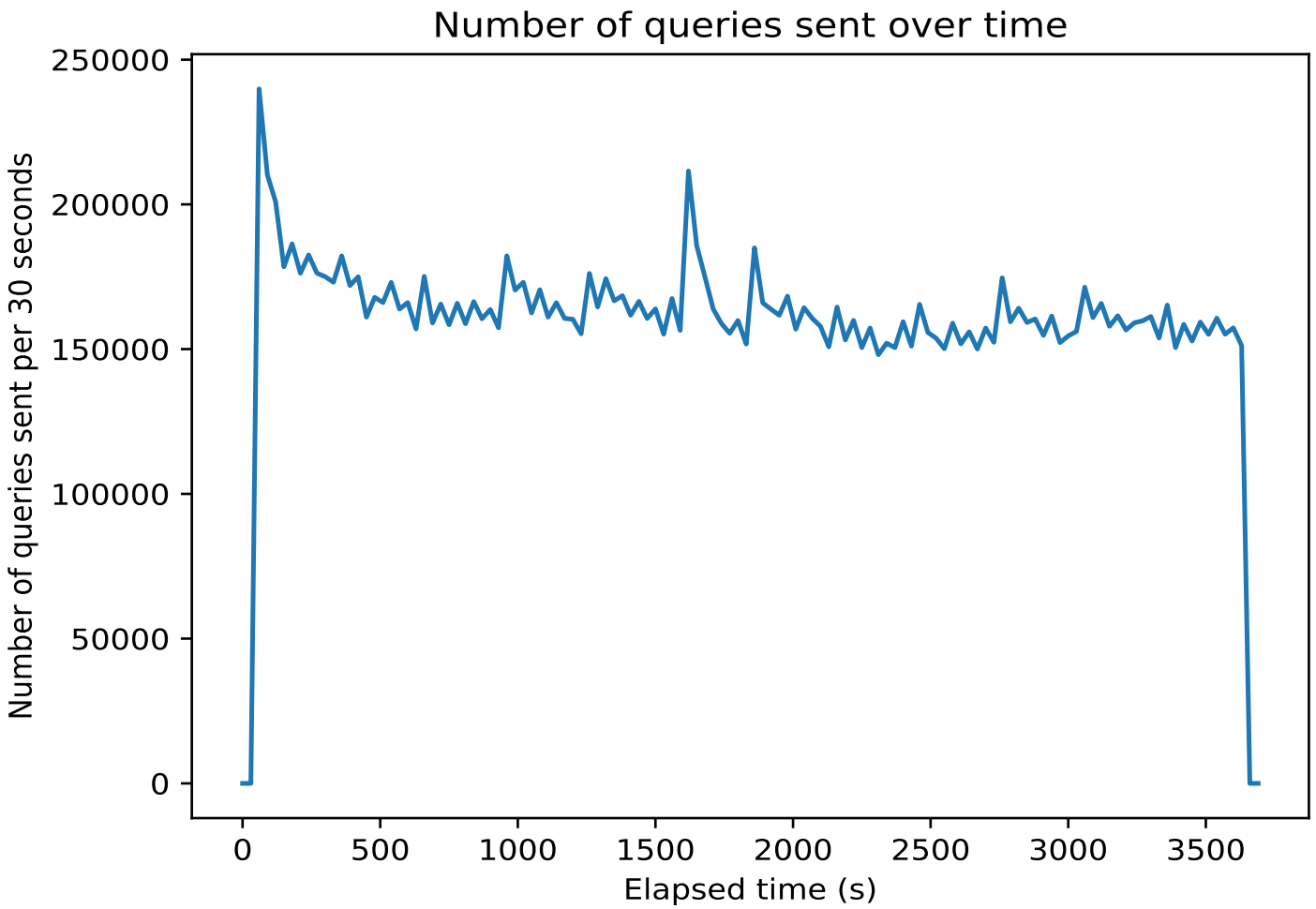


TCP to .nl

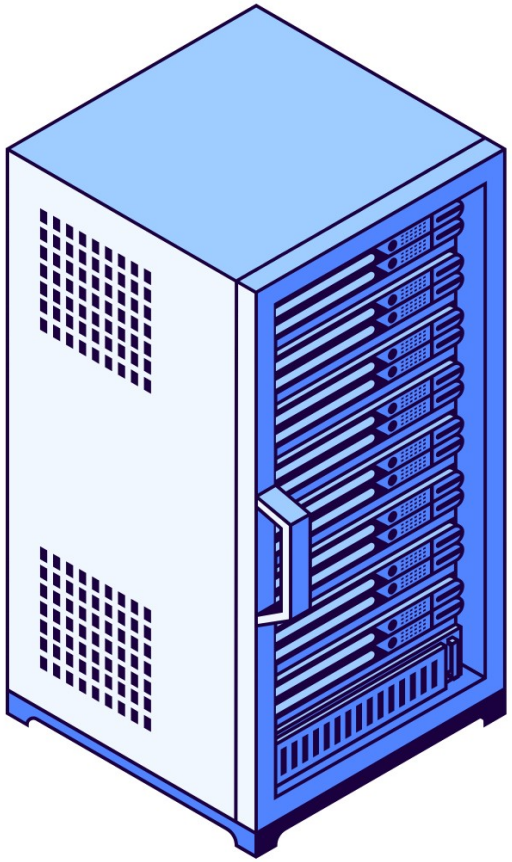


~1% TCP

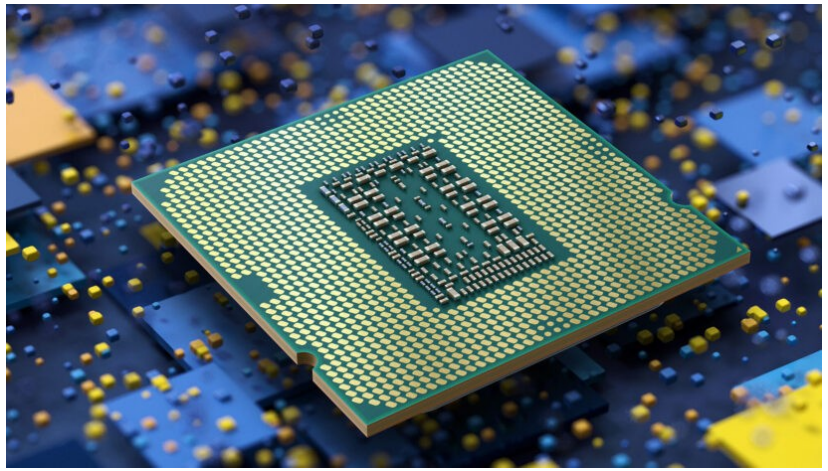
Query data


Number of queries sent over time
Number of queries sent per 30 seconds
250000
200000
150000
100000
50000
0
0
500
1000
1500
2000
2500
3000
3500
Elapsed time (s)

Test setup



Test
server



CPU time

DNS server software

PowerDNS

NSD

KNOT DNS

BIND 9

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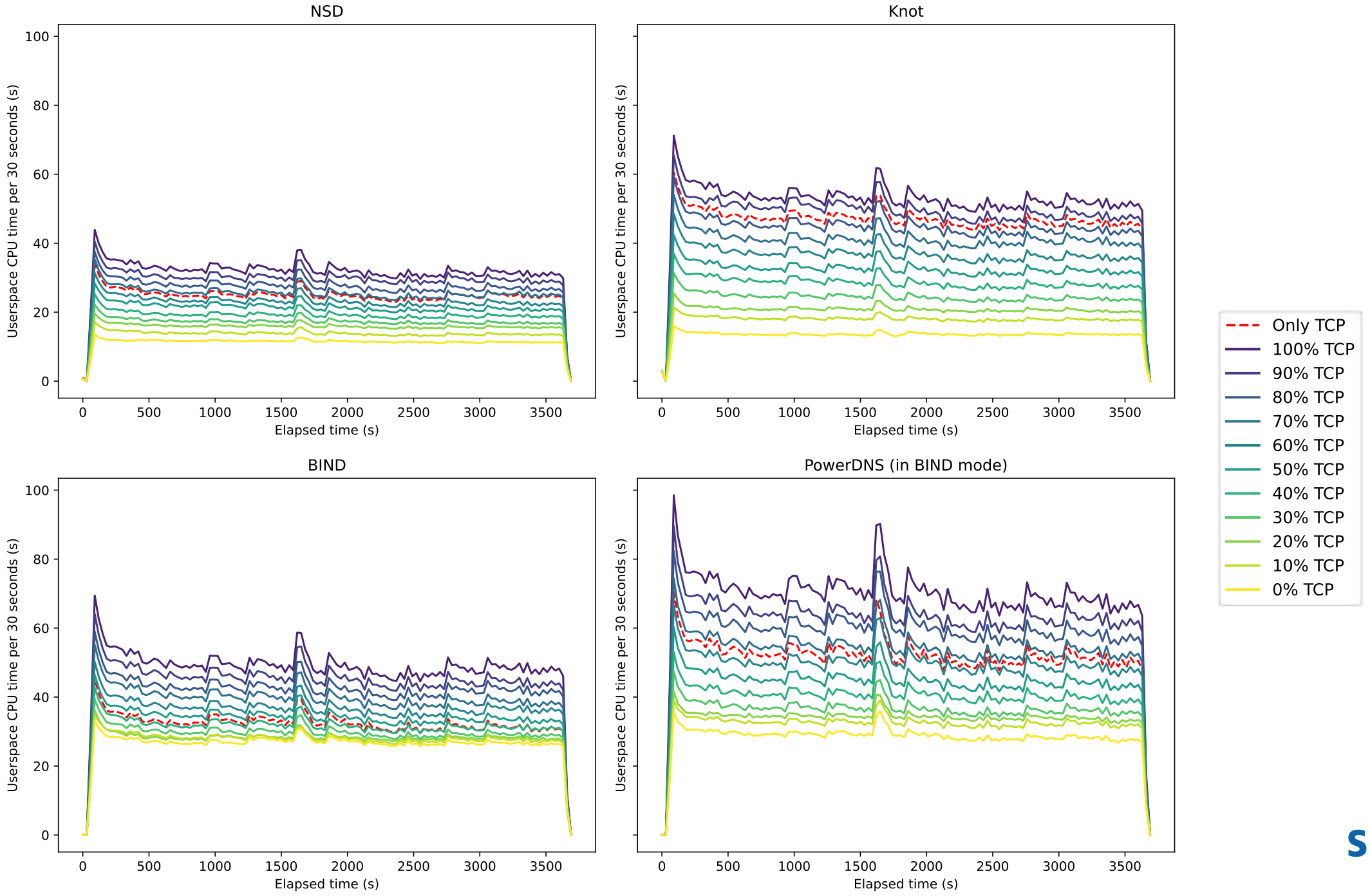
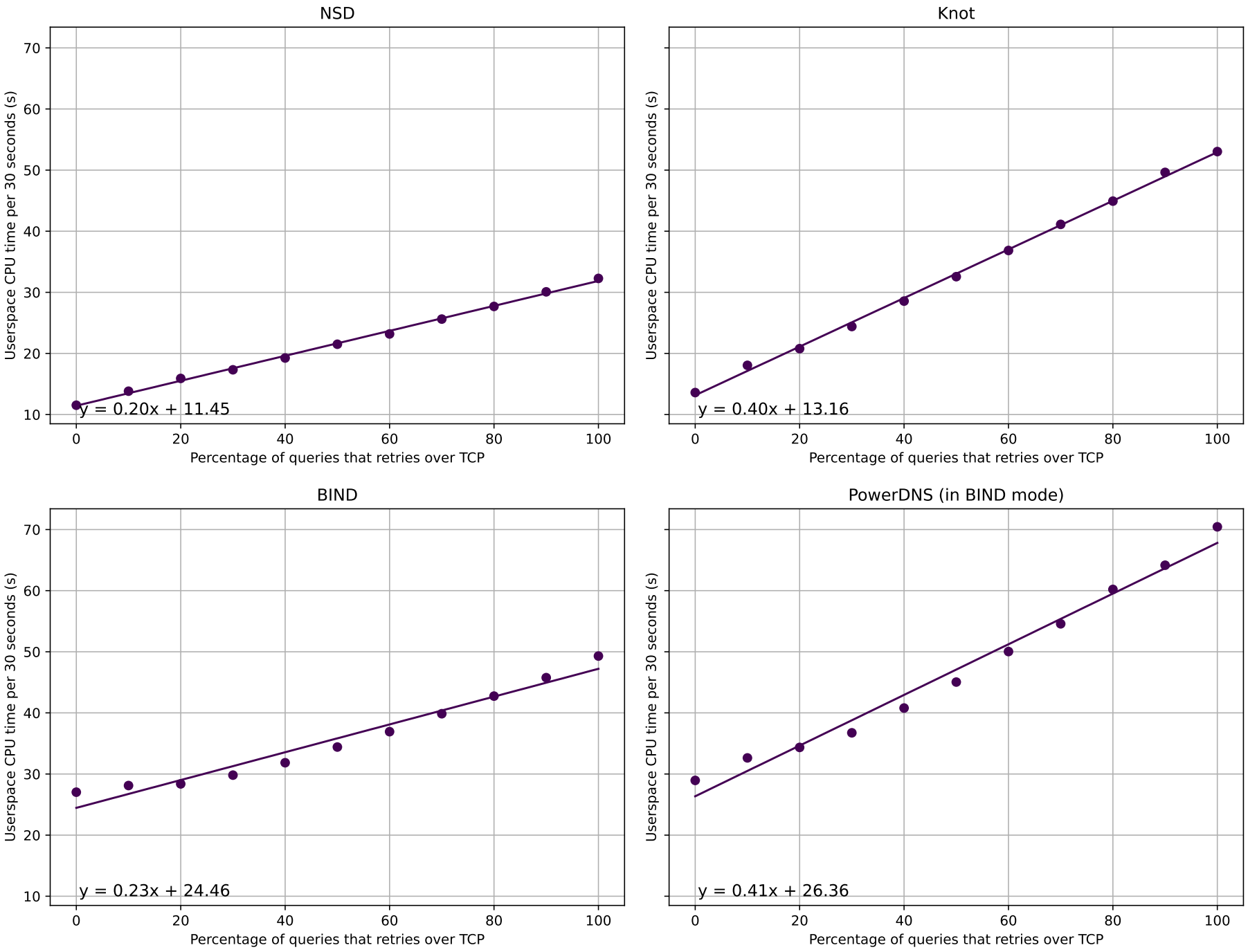
Experiment: percentages of TCP retries

Experiment: only TCP


NSD
Knot
BIND
PowerDNS (in BIND mode)
Userspace CPU time per 30 seconds (s)
Elapsed time (s)
0
20
40
60
80
100
0
500
1000
1500
2000
2500
3000
3500
Only TCP
100% TCP
90% TCP
80% TCP
70% TCP
60% TCP
50% TCP
40% TCP
30% TCP
20% TCP
10% TCP
0% TCP


NSD
Knot
BIND
PowerDNS (in BIND mode)
Userspace CPU time per 30 seconds (s)
Percentage of queries that retries over TCP
y = 0.20x + 11.45
y = 0.40x + 13.16
y = 0.23x + 24.46
y = 0.41x + 26.36
10
20
30
40
50
60
70
0
20
40
60
80
100

Thanks for your attention

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https://www.sidnlabs.nl/en/news-and-blogs/internship-report-dnssec-and-pqc-practical-impact-of-increased-tcp-in-dns